CHRIS FU

Data Engineer | Analytics Engineer

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EXPERIENCE

Senior Data Engineer
Community Federal Savings Bank

Sept 2024 – Present (Full-time) | Remote

- Built the backend data pipeline powering the data quality dashboard, enabling real-time visibility into pipeline health and data integrity
- Designed and implemented Dev, Staging, and Production Astronomer environments while standardizing Snowflake Dev and Staging to mirror Production structure, roles, permissions, and access controls, ensuring consistent deployments and reliable testing across environments
- Migrated the data eng codebase from Bitbucket to GitHub, consolidating 6 repos into a single repo to streamline development and version control
- Led the migration of dbt workloads from dbt Cloud to Astronomer and stood up a dbt documentation website, improving integration, model visibility, and data discoverability
- Worked with Finance to create dbt marts/reports and save 40 man-hours
- Partnered with Compliance and uncovered $30 million in potential fraud

Senior Data Engineer
NationsBenefits

Feb 2023 – Sept 2024 (Full-time) | Remote

- Worked with Finance to improve inefficiency in monthly manual process
- Created dbt macros to assist with developing and testing of 300+ million row models in the dev environment, saving 3+ man-hours per model
- Responsible for dbt transformation uptime in several airflow pipelines
- Create, test, and maintained models in the transformation layer for internal stakeholders to use for ad-hoc reporting and dashboard development
- Parsed JSON blobs in raw to staging models for performance optimization
- Wrote airflow operator to generate .xlsx reports from template and .sql

Data Engineer
Convene

Nov 2020 – Dec 2022 (Full-time) | Remote

- Migrated from Redshift to Snowflake to save compute/storage costs
- Redesigned the data engineering workflow to use Apache Airflow & Docker
- Monitored containers w/ AWS Cloudwatch & ECS to ensure Airflow uptime
- Created a MVP and pitched dbt for data transformation instead of views
- Created an ingestion pipeline to sync data from Salesforce API to Snowflake

Data Engineer
American Airlines

Jan 2020 – Nov 2020 (Full-time) | Fort Worth, TX

- Developed an object-oriented Python script to monitor failures in pipeline
- Re-designed and develop ingestion pipelines from on-prem to cloud
- Developed a PySpark script to ingest AVRO and flatten JSON in Databricks

PROJECTS

Webscrape For Deals

- Built a Python program to search through Macbook Air M1 availability
- Used Selenium to web-scrape through all major US cities

EDUCATION

B.S. in Industrial & Sys Engineering
Georgia Institute of Technology

May 2015 – Dec 2018

CERTIFICATES

Analytics: Essential Tools & Methods
edX

LANGUAGES

- SQL
- Python
- R
- LaTeX
- Java
- HTML & CSS

DATA ENGINEERING

- dbt
- Snowflake
- Microsoft SQL Server
- Airflow
- S3
- Databricks
- Redshift
- Spark
- Docker
- PowerBI

PLATFORM FLAVORS

AWS | Snowflake | Azure Databricks

PROGRAMS & TOOLS

VSCode | DBeaver | Microsoft SQL Studio | Azure DevOps | JIRA | Cloudwatch | ECS | EC2 | Postman | Smartsheets | Github | Jupyter | Tableau | OpenRefine

DATA ENG. CONCEPTS

Primary Keys | Transformation | Ingestion | Modeling | Pipelines | Backfilling | DAG | Monitoring | Validation | Semi-Structured | TaskFlow | Worker | Scheduler | ELT | API